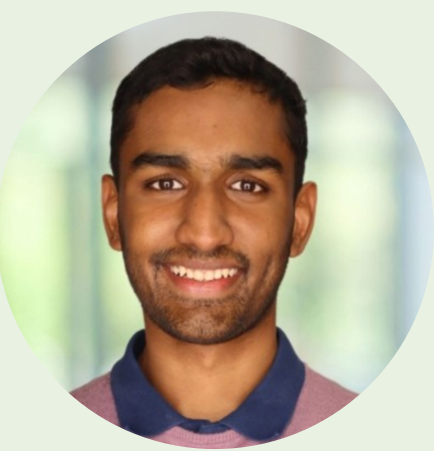




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102

commercial AI agents reviewed

SYSTEMATIC REVIEW CORPUS

31

participants observed

THINK-ALoud STUDY

5

usability barriers identified

THEMATIC ANALYSIS

6

design recommendations

FOR AGENT BUILDERS

1. The state of the world.

- Industry benchmarks test whether agents *can* complete tasks: **capability**, not **usability**.
- They sidestep the human side: **mental models**, **trust**, **oversight**, **recovery**.
- We bridge the gap: audit the marketed promise of **102 commercial agents**, then test whether real end-users can **realize** it on two leading agents.

3. What the industry sells.

102 marketed-as-agentic products cluster into three umbrella use cases. *Categories overlap.*

Orchestration

Act on behalf of users in software interfaces: GUIs, automation.

e.g. Agentforce, Copilot

Creation

Generate structured artifacts: slides, websites, apps, docs.

e.g. Lovable, Gamma, Beautiful.AI

Insight

Distill large information sets into structured user-facing takeaways.

e.g. Perplexity, Spotify AI DJ

31

33

87

4. How we investigated it.

We tested whether participants could realize the marketed promise on two leading commercial agents.

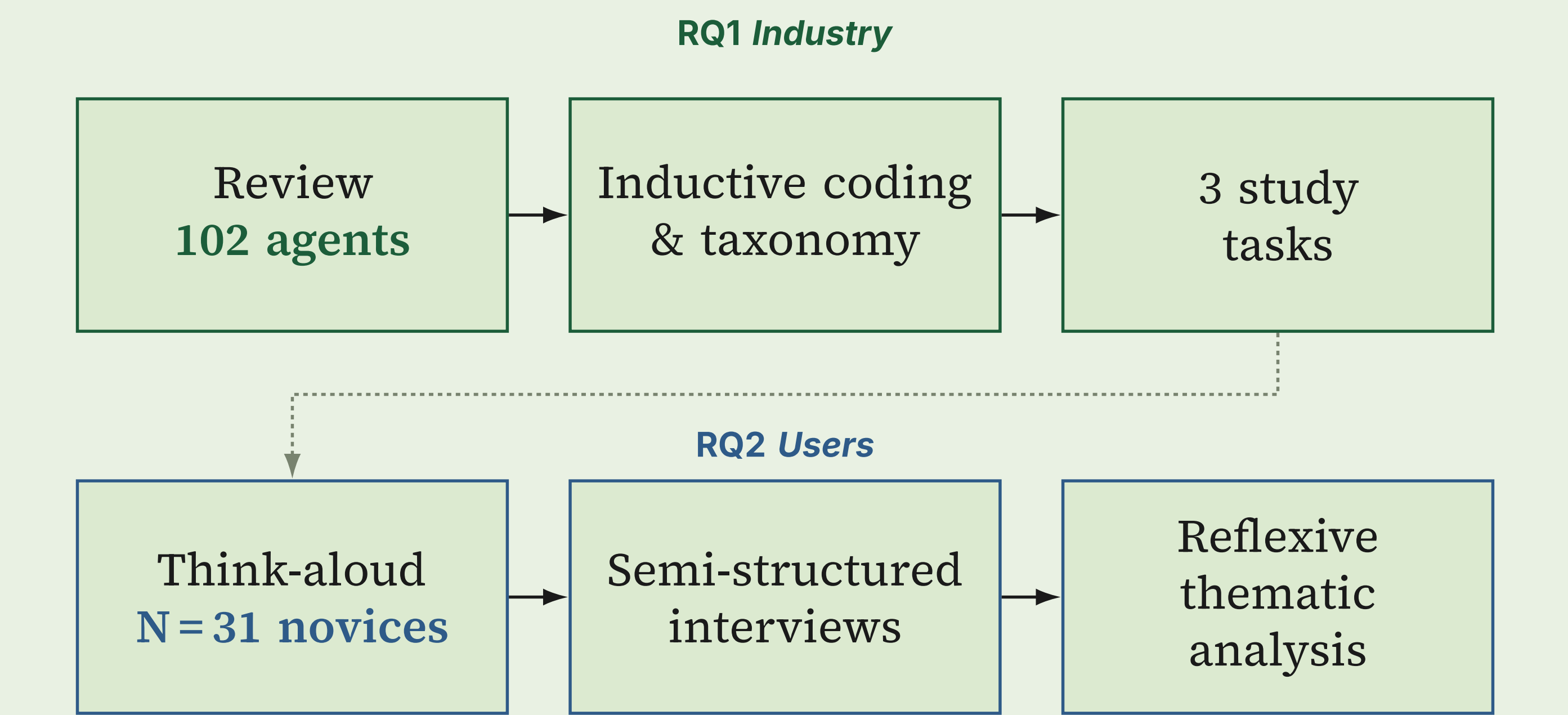


FIG. 1 Mixed-methods pipeline. RQ1 builds the taxonomy; RQ2 tests whether real users realize it. User study approved by Carnegie Mellon University IRB.

5. Quantitative measurements.

Two commercial agents, **Operator** & **Manus**, rated by participants using the System Usability Scale (SUS, 0–100).

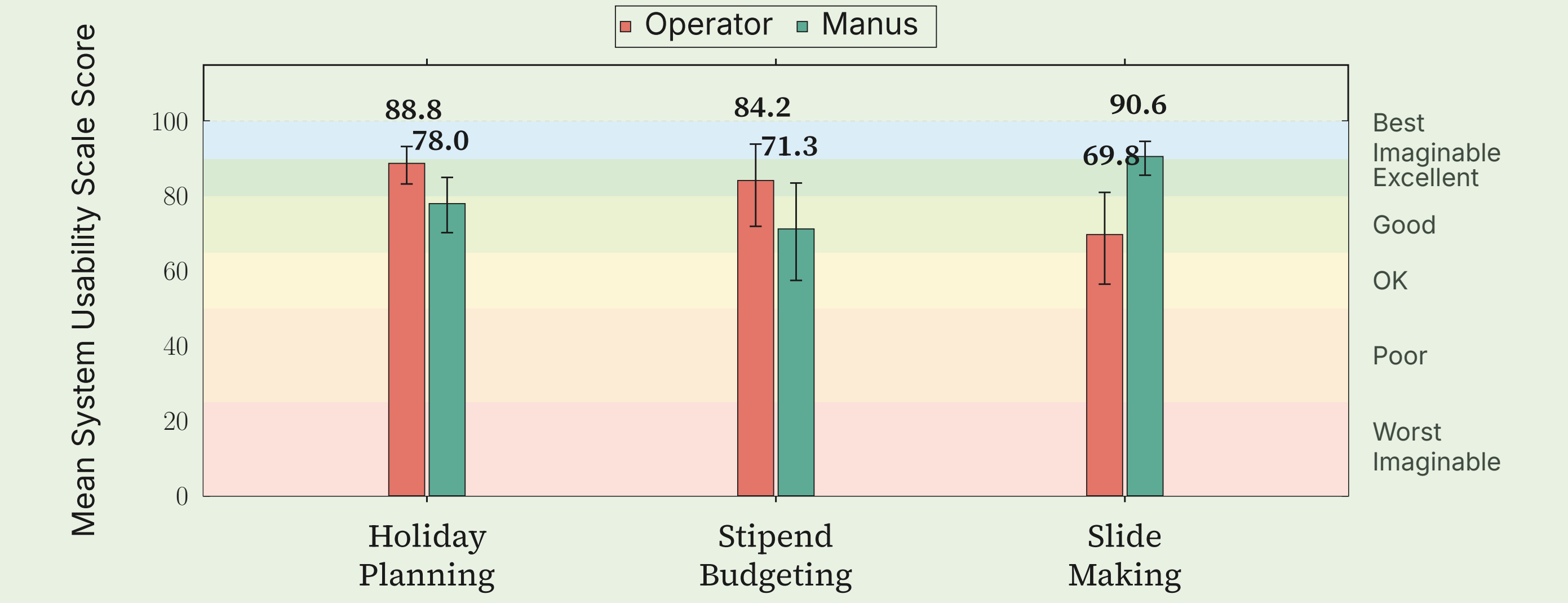


FIG. 2 Mean SUS by task and agent. *n* = 9–13 per cell. Error bars show 95% bootstrap CIs of the mean. Bands are Bangor et al. adjective ratings.

THE SUS PARADOX
Excellent SUS scores, yet five critical usability barriers persist beneath them.

7. Five barriers to agent delegation.

1. Misaligned mental models

Users can’t predict capabilities, turning delegation into frustrating “prompt gambling.”

“You’ve got to sit there and make sure that your initial prompt is perfect.”

2. Presuming trust

Agents expect immediate delegation without first establishing credibility or competence signals.

“I was waiting for it to ask me for some more information.”

3. Inflexible collaboration

Agents act as “lone-wolf” tools that fail to adapt to a user’s need for hands-on oversight.

“It’s kind of like I’m giving you a job, and you’re throwing the job back at me.”

4. Communication overhead

Excessive output and demand for articulation of complex, subjective preferences.

“Oh, my God! It threw out so much stuff... it’s almost overwhelming.”

5. Metacognitive gaps

Agents lack self-awareness of errors, leading to time-wasting “try-fail cycles.”

“It just was kind of circling... seeking to provide an answer rather than to say “I don’t know.””

8. Six design recommendations for building usable next-generation agents.

1. Know your user.

Elicit preferences, skills, and collaboration styles before executing.

2. Know yourself.

Recognise capability limits, stop confidently, surface uncertainty.

3. Be adaptable.

Adapt to task type, confusion signals, and the user’s control preference.

4. Measure twice, cut once.

Surface planning artifacts and mid-task checkpoints to steer before failure.

5. Show, don’t tell.

Support modalities beyond text: examples, sketches, direct selection.

6. Next time’s the charm.

Enable precise contextual iteration, not new prompts that regress elsewhere.

9. Three sociotechnical axes for evaluating agentic systems.

1. Failure recovery.

Intervention frequency, recovery time, and stall rate when stuck.

2. Authorship control.

Plan adherence, precise-edit accuracy, and mid-task checkpoints.

3. Trusted delegation.

Credential management and least-privilege, observable handoffs.